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- IPClinical will inform you with prior notice as it may alter or suspend the IPClinical API or apart or property thereof, or refresh and change these terms and conditions.
- The sole reason for this API will be to acquire the protected health information (PHI) of patients and by accessing or utilizing the material on this API, you agree to be bound by these terms and conditions.

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Fees

IPClinical does not currently charge any fees for the use of the API. IPClinical reserves the right to charge fees for future use of or access to the API. Should we exercise our right to charge fees in the future, we will provide you with advance notice of this decision, and will describe any and all such fees in detailed plain language.

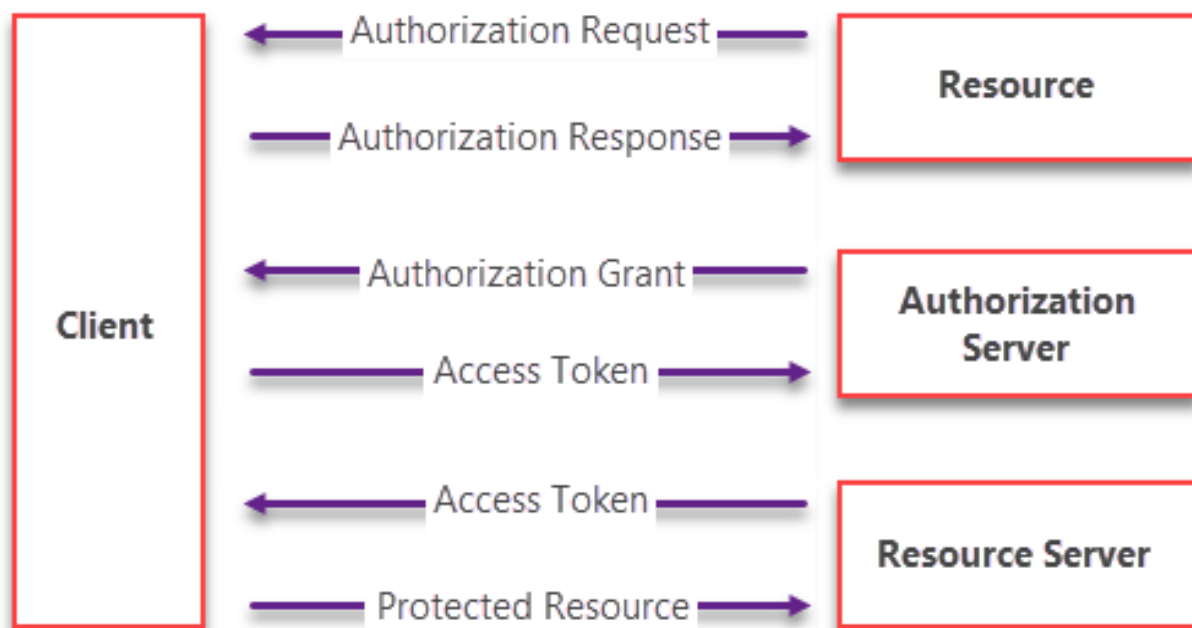


Introduction

This document is created by IPClinical for the clients that use its API to access protected health information (PHI) of the patients. This document provides a comprehensive step by step approach for the users to understand authentication and methods to connect to the IPClinical API and query patient health data.

Authentication Process

IPClinical API uses OAuth2.0 protocol. Please see the chart below:





Generating Authentication Request

Requests can be generated by the authorized users only. Upon successful authorization, a user can access the patient health information from IPClinical. A unique access_token is granted to each user which is valid for a limited time. After the access_token is expired, authorized user will be given a new one. An access_token is a combination of alphanumeric characters which a user requests after they are provided with a client ID and client secret by the IPClinical API.

Request

Please see below the sample URL:

http://192.168.0.18/oauth2/access_token

Following are the parameters required to complete the request for access_token:

client_id	Unique identifier to access the API
client_secret	Secret answer for authentication
grant_type	Value for Client_Credentials

IPClinical will assign client_id and client_secret to its vendors.

Response

Following response will be given as a result of a successful request:

access_token	Token required to access the API
expires_in	Time of the expiry of access token



Patient Search

Once the user has been successfully authorized, they can proceed to access the required patient health information. They then need to access patient search URL where they can enter the granted access_token to complete the request. Without the access_token, request will not be accepted. In order to search a patient, the parameters given below will be entered.

first_name	Patient first name (Required)
last_name	Patient last name (Required)
date_of_birth	Date of birth in YYYY-MM-DD format (Required)
gender	Patient gender (Optional)
phone_number	Phone number of patient (Optional)

Gender field will only accept below values:

- i. "M" for Male
- ii. "F" for Female
- iii. "U" for Unknown

The parameters will be passed in the form of a string. IPClinical returns the response in below format:

```
{
  "patient_id": "S1-S2-Sa-T1-S3-I2-S4-===",
  "first_name": "sample string 1",
  "last_name": "sample string 2",
  "date_of_birth": "sample string 3",
  "gender": "sample string 5",
  "phone_number": "sample string 4",
}
```

If the parameters are not entered in the said format, the request will not be accepted and the user will get an error. In case of a successful request, user will be given a unique id for the selected patient through whom the required health records for that patient can be accessed.



All Patient Data

Once the patient id has been received, user may either get the patient data in categories or the completed data. It is not mandatory to enter the dates when requesting all of the patient data. User can select a date range if they want to. In case a date is not selected, API will return all data automatically. It is mandatory to enter the other date if one of them is selected e.g., end date becomes mandatory if start date is entered and vice versa.

USCDiv1 data classes & elements recognized by the API

DATA CLASS	DATA ELEMENTS
Patient Demographics/Information	Patient First Name, Last Name, Birth Sex, Date of Birth, Race & Ethnicity, Language
Smoking Status	Smoking status
Assessment & Plan of Treatment	Health professional's conclusions and working assumptions that will guide treatment of the patient.
Goals	Desired state to be achieved by a patient.
Care Team Members	Information on a person who participates or is expected to participate in the care of a patient.
Health Concerns	Health related matter that is of interest, importance, or worry to someone who may be the patient, patient's family or patient's health care provider.
Problems	Condition, diagnosis, or reason for seeking medical attention.
Medications	Pharmacologic agents used in the diagnosis, cure, mitigation, treatment, or prevention of disease.
Allergies & intolerances	Harmful or undesired physiological responses associated with exposure to a substance.
Immunizations	Record of vaccine administration.
Laboratory	Analysis of clinical specimens to obtain information about the health of a patient.
Unique Device Identifier(s) for a Patient's Implantable Device(s)	Unique identifier(s) for a patient's implantable device(s).



Procedures	Activity performed for or on a patient as part of the provision of care.
Vital Signs	Physiologic measurements of a patient that indicate the status of the body's life sustaining functions.
Clinical Notes	Narrative patient data relevant to the context identified by note types.
Provenance	The metadata, or extra information about data, regarding who created the data and when it was created.

Request

Please see below the sample URL:

http://192.168.0.18/oauth2/access_token

Header Parameters

Authorization	access_token e.g., Bearer fskdjf36saQ
Content_type	Format e.g., JSON (application/JSON)
Host	Provided URL

Body Parameters

PatientKey	Unique id for the patient
DateRange	Specific date or date range
IncludedComponents	Required components
ResponseType	XML, JSON

Example



```
{
  "PatientKey": "sample string 1"
  "DateRange": {
    "StartDate": "sample string 1",
    "EndDate": "sample string 2"
  },
  "IncludedComponents":{
    "AllDataAccess":true,
  },
  "ResponseType": "sample string 2"
}
```

Response

API returns the patient data as a CCD file in XML format. Result would be returned in JSON body.